

Exercise 3: Google App Engine SDK and Web App Development

Aim

To install the Google App Engine SDK and develop a Hello World / simple web application using Python.

Procedure

Step 1: Install Google Cloud CLI & Python

Ensure Python 3 is installed on the system.

```
ubuntu@ubuntu:~$ sudo apt install python3
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 3
```

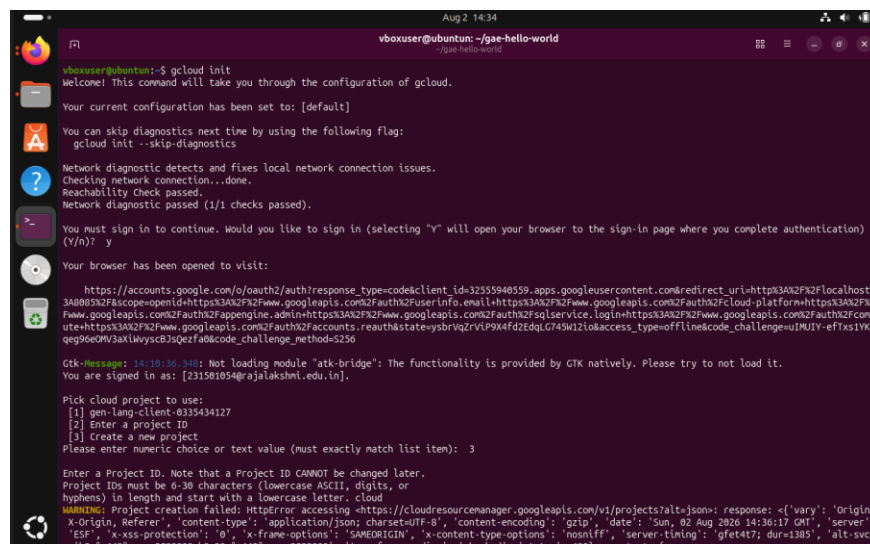
Download and install the Google Cloud CLI (contains the modern App Engine SDK) from cloud.google.com/sdk/docs/install:

```
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli-  
linux-x86_64.tar.gz  
tar -xf google-cloud-cli-linux-x86_64.tar.gz  
./google-cloud-sdk/install.sh
```

Step 2: Initialize GCloud & Install Components

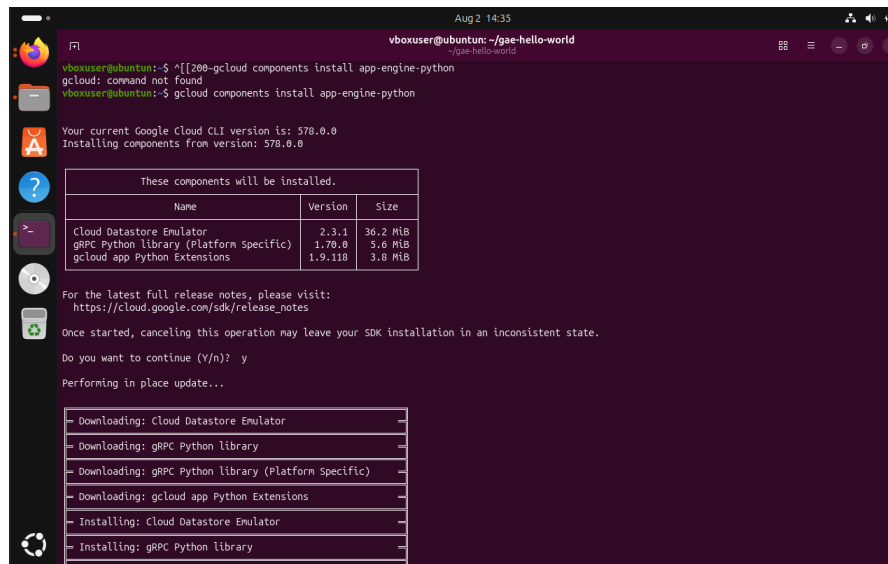
```
gcloud init
```

Sign in to your Google account when prompted.



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gcloud components install app-engine-python



A terminal window titled 'vboxuser@ubuntu: ~/gae-hello-world' showing the execution of 'gcloud components install app-engine-python'. The terminal output indicates the current Google Cloud CLI version is 578.0.0 and that components are being installed from version 578.0.0. A table lists the components to be installed: Cloud Datastore Emulator (2.3.1, 36.2 MiB), gRPC Python library (Platform Specific) (1.78.0, 5.6 MiB), and gcloud app Python Extensions (1.9.118, 3.8 MiB). Below the table, a link to release notes is provided. The user confirms the installation with 'y'. The terminal then shows a progress bar for downloading and installing the components.

```
vboxuser@ubuntu:~/gae-hello-world$ gcloud components install app-engine-python
gcloud: command not found
vboxuser@ubuntu:~/gae-hello-world$ gcloud components install app-engine-python

Your current Google Cloud CLI version is: 578.0.0
Installing components from version: 578.0.0

These components will be installed.


| Name                                    | Version | Size     |
|-----------------------------------------|---------|----------|
| Cloud Datastore Emulator                | 2.3.1   | 36.2 MiB |
| gRPC Python library (Platform Specific) | 1.78.0  | 5.6 MiB  |
| gcloud app Python Extensions            | 1.9.118 | 3.8 MiB  |



For the latest full release notes, please visit:
https://cloud.google.com/sdk/release\_notes

Once started, canceling this operation may leave your SDK installation in an inconsistent state.

Do you want to continue (Y/n)? y
Performing in place update...

- Downloading: Cloud Datastore Emulator
- Downloading: gRPC Python library
- Downloading: gRPC Python library (Platform Specific)
- Downloading: gcloud app Python Extensions
- Installing: Cloud Datastore Emulator
- Installing: gRPC Python library
```

Step 3: Create the App Directory

```
mkdir gae-hello-world
cd gae-hello-world
```

Step 4: Create Application Code (main.py)

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello World from Google App Engine!"

if __name__ == "__main__":
    app.run(host="127.0.0.1", port=8080, debug=True)
```

Step 5: Create Configuration File (app.yaml)

```
runtime: python39
handlers:
- url: /*
  script: auto
```

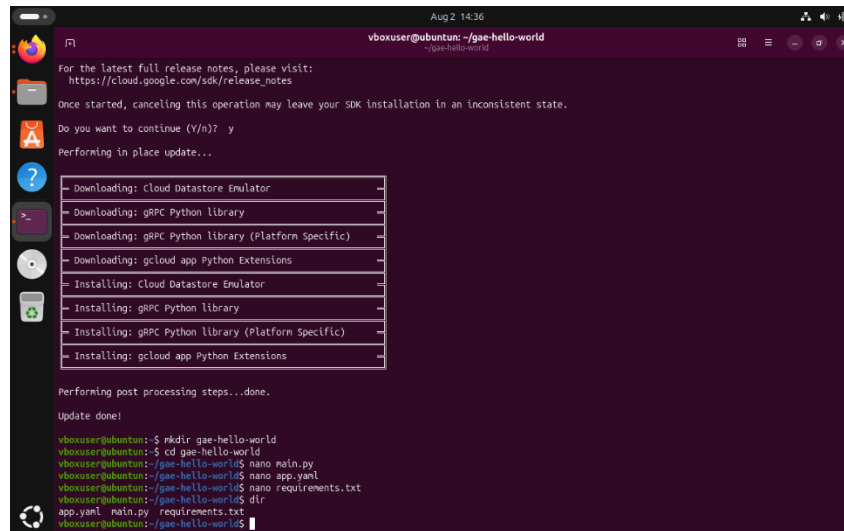
Step 6: Create requirements.txt

```
Flask==2.0.1
```

Output

Files created in project directory (main.py, app.yaml, requirements.txt):

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The image shows a terminal window titled 'vboxuser@ubuntu: ~/gae-hello-world' with a timestamp of 'Aug 2 14:36'. The terminal displays the following text:

```
For the latest full release notes, please visit:
https://cloud.google.com/sdk/release_notes

Once started, cancelling this operation may leave your SDK installation in an inconsistent state.
Do you want to continue (Y/n)? y
Performing in place update...

- Downloading: Cloud Datastore Emulator
- Downloading: gRPC Python library
- Downloading: gRPC Python library (Platform Specific)
- Downloading: gcloud app Python Extensions
- Installing: Cloud Datastore Emulator
- Installing: gRPC Python library
- Installing: gRPC Python library (Platform Specific)
- Installing: gcloud app Python Extensions

Performing post processing steps...done.
Update done!

vboxuser@ubuntu:~/gae-hello-world$ mkdir gae-hello-world
vboxuser@ubuntu:~/gae-hello-world$ cd gae-hello-world
vboxuser@ubuntu:~/gae-hello-world$ nano main.py
vboxuser@ubuntu:~/gae-hello-world$ nano app.yaml
vboxuser@ubuntu:~/gae-hello-world$ nano requirements.txt
vboxuser@ubuntu:~/gae-hello-world$ dir
app.yaml  main.py  requirements.txt
vboxuser@ubuntu:~/gae-hello-world$
```

Result

Thus, the Google App Engine SDK was installed and a simple Hello World web application was developed using Python and Flask.